1. Competency #7: Manipulate numbers, graphical/statistical data, algebraic expressions, equations, and geometric representations

> *Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations*

Highly relevant (⭐⭐⭐⭐⭐)
The core of this problem lies in **numerical manipulation grounded in geometry**:

- Deriving radius from diameter (140 ÷ 2 = 70);
- Subtracting radius from the maximum height (150 − 70 = 80).

This directly demonstrates the integration and manipulation of **geometric concepts and numerical data**, aligning perfectly with Competency #7.

---

2. Competency #13: Use contextual understanding to guide or expedite the mathematical solving process (e.g., choosing appropriate precision)

> *Use an understanding of the context to guide or expedite the mathematical solving process...*

Highly relevant (⭐⭐⭐⭐☆)
Students must understand the **physical structure of a Ferris wheel**:

- The highest point = center height + radius (not diameter);
- The riverbed serves as the reference zero level;
- No need for trigonometry or coordinate systems—the real-world context simplifies the model.

Ignoring the fact that a Ferris wheel is circular could lead to errors like using the full diameter (e.g., 150 − 140 = 10 m), yielding an absurd answer. Thus, **real-world knowledge directly steers the correct mathematical approach**, matching Competency #13.

---

3. Competency #3: Implement a given strategy to determine a mathematical solution

> *Implement a given strategy to determine a mathematical solution*

Moderately relevant (⭐⭐⭐☆☆)
Although no explicit strategy is provided, the standard solution path is well-established: **subtract the radius from the highest point**. Students must **execute this clear, procedural strategy**—a hallmark of Competency #3.

4. Competency #1: Perform a simple calculation

> *Perform a simple calculation*

Moderately relevant (⭐⭐⭐☆☆)
The arithmetic itself is straightforward (150 − 70 = 80), but it is essential to the solution. Even basic computations are part of "employing" mathematics in applied contexts, making Competency #1 applicable.

---

5. Competency #12: Use a multi-step procedure to arrive at a mathematical solution, conclusion, or generalization

> *Use a multi-step procedure leading to a mathematical solution...*

Moderately relevant (⭐⭐⭐☆☆)
The solution involves two distinct steps:

1. Compute radius from diameter (140 ÷ 2 = 70);
2. Subtract radius from maximum height (150 − 70 = 80).

This constitutes a **concise yet complete multi-step procedure**, satisfying the requirements of Competency #12.

6. Competency #5: Understand and apply constructs based on definitions, rules, and formal systems (including familiar algorithms)

> *Understand and utilise constructs based on definitions, rules and formal systems...*

Moderately relevant (⭐⭐⭐☆☆)
Students must invoke the **definition of a circle**: "the radius is the distance from the center to any point on the circumference." This foundational geometric rule is essential to setting up the correct relationship. Applying such a definition reflects Competency #5.

---

7. Competency #2: Select an appropriate strategy from a list

Low relevance (⭐☆☆☆☆)
No list of strategies is provided, and there's no need to choose between multiple approaches (e.g., algebraic vs. geometric). The solution path is unique and direct, so Competency #2 is not significantly engaged.

---

8. Competencies #4, #6, #8–#11, #14: Drawing diagrams, extracting data, articulating reasoning, using tools, switching representations, generalizing, etc.

> *Make diagrams... Extract info from graphs... Articulate solution... Use tech tools... Switch representations... Make generalisations*

Low or not relevant (⭐☆☆☆☆)

- No diagram creation is needed (one is already provided);
- No complex data extraction from charts or graphs;
- No requirement to explain or justify steps (answer is a single number);
- No use of digital tools, representation switching, or generalization.

These competencies are minimally or not at all activated in this problem.

1. 第7条：操作数字、图形、统计信息、代数式、方程和几何表示

> *Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations*

高度相关（⭐⭐⭐⭐⭐）
本题的核心操作是**几何量的数值运算**：

- 从“直径 140 m”推导出“半径 70 m”（除法）；
- 用“最高点 150 m”减去“半径 70 m”得到圆心高度（减法）。

这直接体现了对**几何表示（圆）与数值数据**的联合操作，完全契合第7条。

---

2. 第13条：利用情境理解指导或加速数学求解过程（如采用合适精度）

> *Use an understanding of the context to guide or expedite the mathematical solving process...*

高度相关（⭐⭐⭐⭐☆）
学生必须理解摩天轮的**物理结构**：

- 最高点 = 轮心 + 半径（而非直径）；
- 河床是高度零点；
- 不需要复杂三角或坐标系——情境简化了模型。

若忽略“摩天轮是圆”这一常识，可能误用直径直接计算（如 150 − 140 = 10 m），得出荒谬结果。因此，**现实情境知识直接指导了正确的数学路径**，符合第13条。

---

3. 第3条：实施给定策略以确定数学解

> *Implement a given strategy to determine a mathematical solution*

中等相关（⭐⭐⭐☆☆）
虽然题目未明确“给定策略”，但标准解法是公认的：**用最高点减去半径**。学生需**执行这一清晰、直接的程序性策略**。这属于“实施已知算法/步骤”的范畴，符合第3条。

4. 第1条：进行简单计算

> *Perform a simple calculation*

中等相关（⭐⭐⭐☆☆）
计算本身极其简单（150 − 70 = 80），但它是解题的必要环节。尽管看似基础，但**准确执行基本算术**仍是“运用”能力的一部分，尤其在应用题中。因此第1条适用。

---

5. 第12条：使用多步骤程序得出数学解、结论或推广

> *Use a multi-step procedure leading to a mathematical solution...*

中等相关（⭐⭐⭐☆☆）
解题包含两个明确步骤：

1. 由直径求半径（140 ÷ 2 = 70）；
2. 用最高点减半径（150 − 70 = 80）。

这是一个**简洁但完整的多步程序**，符合第12条对“过程性求解”的要求。

6. 第5条：理解并运用基于定义、规则和形式系统的结构（包括熟悉算法）

> *Understand and utilise constructs based on definitions, rules and formal systems...*

中等相关（⭐⭐⭐☆☆）
学生需调用**圆的基本定义**：“半径是从圆心到圆周任意点的距离”。这是几何形式系统中的核心规则。正确应用此定义是解题前提，体现了对**数学概念规则的运用**，符合第5条。

---

7. 第2条：从列表中选择合适策略

> *Select an appropriate strategy from a list*

低相关（⭐☆☆☆☆）
题目未提供策略选项，也无需在多种方法间权衡（如代数法 vs 几何法）。解法唯一且直接，故第2条不显著。

---

8. 第4、6、8、9、10、11、14条：作图、提取信息、展示过程、使用工具、多表征、推广等

> *Make diagrams... Extract info from graphs... Articulate solution... Use tech tools... Switch representations... Make generalisations*

低或不相关（⭐☆☆☆☆）

- 无需作图（图已给出）；
- 无需提取复杂图表信息（仅读取标注数值）；
- 无需展示中间步骤（答案为单一数值）；
- 未涉及技术工具、多表征转换或一般化。

这些能力在本题中几乎未被调动。